Analytical Angles Dictionary & User Guide (Distilled)

A conceptual menu of analysis patterns for a notebook-executing agent. Written to be general, reusable, and semi-taxonomical. Each entry assumes dataframes and schema context are provided, and the agent will produce Markdown explanations plus executable code that saves artifacts (tables/plots) for rendering.

How to use this guide

- **Pick an angle** (Descriptive, Diagnostic, Predictive, Prescriptive) that matches the user's intent.
- **Combine primitives** (entities, measures, operators, windows) to assemble the analysis.
- **Output**: concise narrative → supporting table/visual → saved artifacts (`plots/…`, `tables/…`).
- **Iterate**: refine with follow-ups (drilldowns, segmentation, sensitivity checks).

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0) Dimensional Grammar (the building blocks)

Entities: time → {hour, day, week, month, quarter, year}; product → {sku, subcategory, category, brand}; geography → {store, region}; channel → {store, ecommerce, bopis}; customer → {id, segment, cohort}; supplier.

Measures (examples): `sales`, `units`, `gross_profit`, `margin_pct`, `aov`, `conversion_rate`, `inventory_on_hand`, `days_of_supply`, `return_rate`, `fill_rate`, `turnover`, `retention_rate`, `gmroi`.

Operators: filter, group, aggregate, rank, normalize, share-of-total, difference, `%chg`, rate, ratio.

Time windows: aligned periods (YoY/QoQ/MoM), rolling windows (e.g., 7/28/90-day), lags/leads for comparisons, backtesting folds for forecasts.

Data splits: time-aware splits for temporal data; stratified splits where class imbalance exists; control/treatment partitions for interventions.

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1) Descriptive Angles ("what happened?")

1.1 Trend Profiling

Intent: reveal level + direction + variability over time. **Core**: resample → aggregate → visualize (line/area). Align comparable periods for PoP/YoY. **Outputs**: trend table; plot with markers for peaks/dips; optional CI bands for smoothed lines. **Edge notes**: align calendars (fiscal vs calendar), handle partial periods.

1.2 Mix & Contribution

Intent: who/what contributed how much. **Core**: compute share-of-total by dimension; rank; 100% stacked charts. **Outputs**: contribution table; Pareto breakdown (A/B/C buckets). **Edge notes**: mix shift can mask true growth—pair with absolute deltas.

1.3 Period-Over-Period Comparison

Intent: compare current vs prior aligned period. **Core**: join aligned periods → compute `abs_delta` and `%chg`. **Outputs**: variance table; waterfall or delta bar chart. **Edge notes**: partial weeks/months; moving holidays.

1.4 Seasonality Map

Intent: detect seasonal structure across dimensions. **Core**: pivot (month × region/category) → heatmap. **Outputs**: heatmap; seasonality index table. **Edge notes**: decompose confounded trend vs season (`level`, `trend`, `seasonal`).

1.5 Productivity Ratios

Intent: normalize by resource (space, labor, traffic). **Core**: compute `sales_per_sqft`, `sales_per_employee`, `conversion_rate`. **Outputs**: ratio league table; scatter (size vs sales; color by region). **Edge notes**: ensure denominators are consistent by period.

1.6 Benchmarking vs Plan/Target

Intent: attainment vs goals/forecast. **Core**: `attainment = actual / target`; highlight gaps; rank over/under achievers. **Outputs**: attainment table; bullet charts. **Edge notes**: reconcile changing targets; document assumption lineage.

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2) Diagnostic Angles ("why did it happen?")

2.1 Variance Decomposition (Driver Tree)

Intent: explain delta using factors (price × volume, mix, availability, etc.). **Core**: factorize change into additive/multiplicative components; Shapley-style or sequential. **Outputs**: driver table; waterfall chart by driver. **Edge notes**: order dependence—state method; check interactions.

2.2 Segmentation & Cohorts

Intent: group by behavior/time of entry to isolate patterns (e.g., retention). **Core**: cohort index (period since first event) → retention/monetization curves; segment by attributes. **Outputs**: cohort matrix; survival/retention plot. **Edge notes**: censoring; seasonality impacts on cohorts.

2.3 Intervention/Lift Analysis

Intent: effect of promos, remodels, events, policy changes. **Core**: pre/post comparison with controls (matched units or regions); diff-in-diff where suitable. **Outputs**: lift % with CI; timeline plot; sensitivity checks. **Edge notes**: parallel trends assumption; spillovers and timing.

2.4 Stockouts, Returns, and Availability

Intent: quantify demand lost and quality issues. **Core**: detect zero-inventory intervals; estimate unmet demand (velocity × outage); compute `return_rate` and reasons if available. **Outputs**: lost-sales estimate; heatmap of stockout frequency; returns leaderboard. **Edge notes**: censoring bias; lead-time alignment.

2.5 Funnel Drop-Off (Digital or Store)

Intent: locate leaks in conversion paths. **Core**: stage counts → stage-to-stage rates → cumulative conversion; segment by device/channel. **Outputs**: funnel table; bar step chart; abandonment summary. **Edge notes**: de-dupe sessions/users; attribute multi-session journeys carefully.

2.6 Elasticity & Sensitivity Scans

Intent: response of demand to drivers (price, availability, rating, promo intensity). **Core**: partial correlations/ regression on panels; local elasticities via small perturbations. **Outputs**: driver coefficients/elasticities; ICE/ PDP visuals (when modeled). **Edge notes**: endogeneity; multicollinearity (check VIF/regularize).

2.7 Cannibalization & Halo

Intent: substitution or complementarity between items after assortment changes. **Core**: pre/post with controls; cross-price/co-movement analysis; basket co-occurrence change. **Outputs**: impact table on incumbents; net category change. **Edge notes**: seasonality confounds; launch phasing.

2.8 Anomaly & Quality Checks

Intent: detect spikes/drops and data issues worth investigation. **Core**: robust z-scores; rolling median/MAD; model-based outlier flags. **Outputs**: anomaly list with context; small multiples of flagged series. **Edge notes**: annotate with events; guard against alert fatigue.

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3) Predictive Angles ("what will happen?")

3.1 Forecasting (Univariate + Exogenous)

Intent: short- to mid-term projections with uncertainty. **Core**: seasonal decomposition; ARIMA/SARIMAX/ ETS; backtest with `TimeSeriesSplit`; include exogenous signals when justified. **Outputs**: forecast table; plot with prediction intervals; backtest error summary. **Edge notes**: leakage avoidance; holiday/fixture calendars; scale per segment when heterogeneous.

3.2 Propensity & Classification

Intent: probability of an event (purchase, churn, return). **Core**: baseline features (recency/frequency/monetary, engagement, availability); calibrated classifiers; cost-aware thresholds. **Outputs**: ROC/PR metrics; calibration curve; threshold-policy table. **Edge notes**: class imbalance; temporal validation; actionability of features.

3.3 Regression for Drivers & Uplift

Intent: quantify effect sizes on continuous outcomes; estimate incremental impact. **Core**: regularized linear models; panel/FE where appropriate; uplift via interaction terms or treatment models. **Outputs**: coefficients with CIs; partial dependence; uplift ranking. **Edge notes**: interpretation vs prediction tradeoffs; clustered/robust SEs.

3.4 Clustering & Profiling

Intent: discover groups among stores/products/customers. **Core**: scale/select stable features; choose `k` via elbow/silhouette; profile each cluster. **Outputs**: cluster membership table; centroid profile cards; scatter embeddings. **Edge notes**: stability across seeds/time; operational labels not just numbers.

3.5 Anomaly Detection (Predictive)

Intent: find rare events or drift proactively. **Core**: isolation forests/LOF on engineered features; PSI/KS for drift between train/serve. **Outputs**: ranked anomalies; drift dashboard with top shifting features. **Edge notes**: feedback loop to retraining; human review thresholds.

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4) Prescriptive Angles ("what should we do?")

4.1 Inventory Readiness

Intent: right-size stock to demand. **Core**: `days_of_supply = on_hand / avg_daily_units`; reorder heuristics from lead time + safety stock; prioritize A-items. **Outputs**: replenishment shortlist; overstocks/understocks heatmap. **Edge notes**: demand volatility; supplier reliability.

4.2 Pricing & Markdown Windows

Intent: tune prices/discounts for revenue/profit goals. **Core**: use elasticities + stock positions; simulate candidate price ladders and expected outcomes. **Outputs**: price/volume/profit scenarios; clearance effectiveness review. **Edge notes**: cross-effects with promos; floor/ceiling constraints.

4.3 Assortment & Discontinuation

Intent: curate portfolio for value. **Core**: ABC + GMROI + trend momentum; flag low-velocity, high-holding-cost items. **Outputs**: keep/grow/prune taxonomy; substitution suggestions. **Edge notes**: halo/cannibalization impacts on neighbors.

4.4 Staffing & Operations Alignment

Intent: match capacity to demand peaks. **Core**: hour/day patterns vs labor schedules; service-level targets. **Outputs**: peak windows and coverage gaps; shift guidance. **Edge notes**: seasonality; special events.

4.5 Promotion & Campaign Optimization

Intent: focus spend where lift is highest. **Core**: prior lift curves + propensity segments; allocate to high-uplift pockets; define holdout tests. **Outputs**: promo calendar with expected lift; post-hoc ROI diagnostics. **Edge notes**: saturation; fairness and guardrails.

4.6 Omnichannel Orchestration

Intent: balance online/store flows. **Core**: BOPIS share trends; ship-from-store loads; latency vs conversion impacts. **Outputs**: regional allocation plan; SLAs and buffer stock by node. **Edge notes**: last-mile costs; substitution policies.

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5) Reusable Patterns & Formulas (quick reference)

- **Growth**: `%chg = (curr - prev) / |prev|` (report sign and base).
- **Contribution**: `share = part / total`; **mix shift** = change in share × level.
- **Elasticity (local)**: `E ≈ (ΔQ/Q) / (ΔP/P)` over small, aligned windows.
- **ABC**: rank by value; cut at ~80/95 cumulative share for A/B (tune to context).
- **Coverage**: `days_of_supply = on_hand / avg_daily_units`; compute per node and global.
- **Returns**: `return_rate = returns / units_sold` (pair with reasons if present).
- **Fill**: `fill_rate = fulfilled / requested` (order or line basis—state which).
- **Retention**: cohort `% active` by age period; repeat rate = returning / prior-year customers.
- **GMROI**: `gross_margin / avg_inventory_cost` (per time unit stated).
- **Promo lift**: `lift = (test - control) / control` on aligned windows; include CI.
- **Backtest**: rolling origin; report `MAE/RMSE/MAPE` with central tendency + dispersion.
- **Thresholding**: choose classifier threshold by business cost/benefit table, not accuracy alone.

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6) Visualization Menu (intent → chart)

- **Trends** → line/area with intervals; annotate events.
- **Comparisons** → grouped bars; 100% stacked for mix.
- **Seasonality** → month×dimension heatmap.
- **Ranking** → horizontal bars; Pareto curve (cum. share).
- **Funnel** → step bars with stage rates.
- **Drivers** → waterfall (variance), scatter with fit line, PDP/ICE for models.
- **Clusters** → scatter in 2D embedding; centroid radar/profile cards.

Conventions: labeled axes with units; clear legends; consistent date formats; save as `plots/<slug>_<yyyy-mm-dd>.png`.

7) Notebook Execution Pattern (thin, for consistency)

1) **Declare intent** in 1–2 lines; name entities, measures, periods, comparison base. 2) **Assemble data**: filters, joins, typing; print shape and null summary. 3) **Compute**: minimal, composable steps; protect against leakage in any modeling. 4) **Validate**: sanity checks, alignment of periods, denominators. 5) **Visualize**: one chart per intent; persist images/tables to paths. 6) **Conclude**: answer plainly; list 1–3 implications or next actions.

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8) Quality & Governance (minimal but firm)

- Use time-aware CV; log seeds/params; prefer simple explanations first.
- Report uncertainty (CIs, backtest errors) alongside point estimates.
- Call out assumptions and notable limitations; avoid over-claiming causality.

Endnotes

This guide is intentionally **concept-first**. It abstracts from domain specifics while staying faithful to common business analytics tasks across sales, inventory, stores, ecommerce, product, customers, and suppliers. Compose complex answers by chaining angles: e.g., *Descriptive trend* → *Diagnostic variance* → *Predictive forecast* → *Prescriptive allocation*. Persist artifacts at each step for transparent, reproducible decision support.